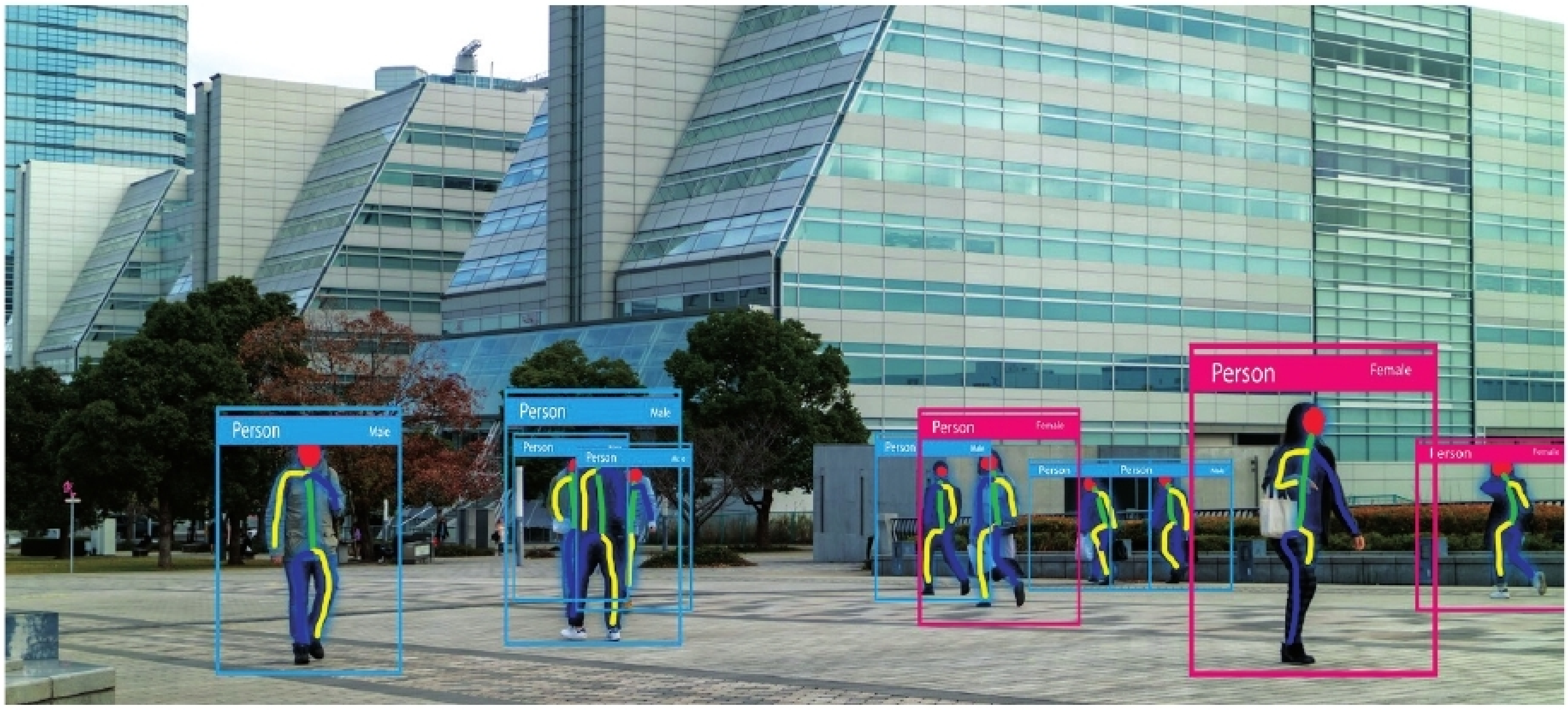
How TensorFlow Can Be Used for Video Analytics

Object detection now plays a very important role in our lives, right from face detection and unlocking a smartphone to detecting bombs in places where people congregate, like airports, bus terminals, railway stations, etc. Such advanced features are a result of the application of machine learning and artificial intelligence. This article discovers how TensorFlow, a machine learning library, can be used to identify objects.



Security and surveillance cameras are very widely used across organisations, so it is important to enhance their effectiveness while saving manpower costs and preventing human errors. In such scenarios, image/video analytics plays a very important role in performing real-time event detection, post-event analysis, and the extraction of statistical and operational data from the videos. Video analytics (VA) is the general analysis of video images to recognise unusual or potentially dangerous behaviour and events in real-time. It can perform three major tasks — provide information, offer assistance, and generate alerts.

Video analytics is widely used for suspicious object detection in schools, in the banking and financial sector, and in critical infrastructure protection. It is also used at heritage sites, commercial spaces, offices, government buildings, factories, airports, railway/metro stations, busy traffic intersections, stadiums, etc. Video analytics is used for monitoring people and vehicle flows. It can spot and prevent the crossing of the speed threshold and vehicular movement in the wrong direction. Other capabilities include crowd counting, incident detection, licence plate recognition, safety alerts, people/object recognition, post-event analysis, lost object detection, etc.

Wikipedia defines many functional applications of video analytics, which are listed in Table 1.

One of the major applications of video analytics is object identification, which is the process of identifying objects within images or videos. It is based on the concept that each object has unique features. These features help to classify the objects into different classes. Identification can be done either by using the machine learning (ML) approach or the deep learning (DL) method.

In the machine learning method, first the features are identified and then support vector techniques (SVM) are used for classification. Deep learning methods are based on convolution neural networks (CNN), which is a kind of neural network that has an input layer, output layer and multiple hidden layers. It uses mathematical model convolution to pass on the result to various successive layers. In this article, I will explain the process of object identification using TensorFlow, with a focus on deep learning methods.